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Tactile feedback mechanism

Abstract

A tactile feedback mechanism is provided, including a movable portion, a fixed portion, a driving assembly, and a connecting assembly. The movable portion is movable relative to the fixed portion. The driving assembly is configured to drive the movable portion to move relative to the fixed portion. The movable portion is movably connected to the fixed portion via the connecting assembly. The connecting assembly is made of a metal material.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/084,312, filed Sep. 28, 2020, the entirety of which are incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present disclosure relates to a tactile feedback mechanism, and more particularly to a tactile feedback mechanism including a connecting assembly.

Description of the Related Art

(2) Many electronic devices (for example, mobile phones, tablets, etc.) used nowadays, generally include a tactile feedback mechanism that generates vibrations, to respond to the operation made by

the user on the electronic device. Many tactile feedback mechanisms in the prior art used eccentric rotating mass (ERM) motors. However, since its vibration is generated by the rotation of the eccentric mass (rotor), it is not conducive to the miniaturization of the mechanism, and it is not easy to accurately control it.

(3) In this regard, the present disclosure provide a tactile feedback mechanism based on a linear resonance actuator (LRA) to generate vibrations. Compared with the tactile feedback mechanism used in the prior art, the tactile feedback mechanism improves in the vibration frequency, vibration intensity, and systematic evaluation.

BRIEF SUMMARY OF THE INVENTION

(4) The present invention provides a tactile feedback mechanism, including a movable portion, a fixed portion, a driving assembly, and a connecting assembly. The movable portion is movable relative to the fixed portion. The driving assembly is configured to drive the movable portion to move relative to the fixed portion. The movable portion is movably connected to the fixed portion via the connecting assembly. The connecting assembly has a metal material.

(5) According to some embodiments of the present disclosure, the connecting assembly includes a first fixed end for the movable portion, a first fixed end for the fixed portion, a first elastic portion. The first fixed end for the movable portion is fixedly connected to the movable portion. The first fixed end for the fixed portion is fixedly connected to the fixed portion. The first fixed end for the movable portion is connected to the first fixed end for the fixed portion via the first elastic portion. The first elastic portion has a metal material. The movable portion has a metal material. The metal material of the first elastic portion is different from the metal material of the movable portion. The density of the metal material of the first elastic portion is lower than the density of the metal material of the movable portion.

(6) According to some embodiments of the present disclosure, the connecting assembly further includes a second fixed end for the movable portion and a second elastic portion. The second fixed end for the movable portion is fixedly connected to the movable portion. The second fixed end for the movable portion is connected to the first fixed end for the fixed portion via the second elastic portion. The first fixed end for the movable portion is connected to the second fixed end for the movable portion via the first fixed end for the fixed portion. The first fixed end for the movable portion, the first fixed end for the fixed portion, and the second fixed end for the movable portion are formed in one piece.

(7) According to some embodiments of the present disclosure, the fixed portion includes a first sidewall. The first fixed end for the fixed portion is disposed on the first sidewall. The movable portion includes a first stopper portion for limiting the range of movement of the movable portion relative to the fixed portion. When the movable portion is in a first limiting position, the first stopper portion directly or indirectly contacts the first fixed end for the fixed portion.

(8) According to some embodiments of the present disclosure, the tactile feedback mechanism further includes a first buffer component. The first buffer component has a plastic material. The first buffer component is fixedly disposed on the first stopper portion or the first fixed end for the fixed portion.

(9) According to some embodiments of the present disclosure, the movable portion includes a first connecting portion and a second connecting portion. The first fixed end for the movable portion is fixed to the first connecting portion. The second fixed end for the movable portion is fixed to the second connecting portion. The first connecting portion and the second connecting portion both have a recessed structure.

(10) According to some embodiments of the present disclosure, the first connecting portion includes a first connecting surface. The second connecting portion includes a second connecting surface. The first fixed end for the movable portion is disposed on the first connecting surface. The second fixed end for the movable portion is disposed on the second connecting surface. The fixed portion includes a second sidewall, the first fixed end for the movable portion and the second fixed

end for the movable portion are arranged along a first axis. The first connecting surface and the second connecting surface face the second sidewall.

(11) According to some embodiments of the present disclosure, the connecting assembly further includes a second fixed end for the fixed portion and a second elastic portion. The second fixed end for the fixed portion is fixedly connected to the fixed portion. The first fixed end for the movable portion is connected to the second fixed end for the fixed portion via the second elastic portion. The first fixed end for the fixed portion is connected to the second fixed end for the fixed portion via the first fixed end for the movable portion. The first fixed end for the movable portion, the first fixed end for the fixed portion, and the second fixed end for the fixed portion are formed in one piece. The first fixed end for the fixed portion and the second fixed end for the fixed portion are arranged along a first axis.

(12) According to some embodiments of the present disclosure, the fixed portion includes a first sidewall. The first fixed end for the fixed portion and the second fixed end for the fixed portion are disposed on the first sidewall. The first fixed end for the movable portion, the first fixed end for the fixed portion, the second fixed end for the fixed portion, and the first sidewall have plate-shaped structures. The first fixed end for the movable portion is located between the movable portion and the first sidewall.

(13) According to some embodiments of the present disclosure, the tactile feedback mechanism further includes a damping component. The movable portion is movably connected to the fixed portion via the damping component. The damping component is flexible, and the damping component has a plastic material, the elastic coefficient of the damping component is lower than the elastic coefficient of the first elastic portion.

(14) According to some embodiments of the present disclosure, the movable portion includes a second stopper portion for limiting the range of movement of the movable portion relative to the fixed portion. When the movable portion is in a second limiting position, the second stopper portion directly or indirectly contacts the second sidewall of the fixed portion.

(15) According to some embodiments of the present disclosure, the tactile feedback mechanism further includes a second buffer component, the second buffer component has a plastic material. The second buffer component is fixedly disposed on the second stopper portion or the second sidewall.

(16) According to some embodiments of the present disclosure, when the first fixed end for the movable portion directly or indirectly contacts both the first fixed end for the fixed portion and the second fixed end for the fixed portion, the range of movement of the movable portion relative to the fixed portion is limited.

(17) According to some embodiments of the present disclosure, the tactile feedback mechanism further includes a first buffer component. The first buffer component has a plastic material. The first buffer component is fixedly disposed on the first fixed end for the movable portion or both the first fixed end for the fixed portion and the second fixed end for the fixed portion.

(18) According to some embodiments of the present disclosure, when viewed along the direction of the main axis, the first elastic portion and the second elastic portion have a pointed shape. The first elastic portion and the second elastic portion are mirror-symmetrical with each other relative to the movable portion.

(19) According to some embodiments of the present disclosure, the connecting assembly further includes an adjusting component. The adjusting component has a plate-like structure, and is disposed on the first elastic portion for adjusting the elastic coefficient of the first elastic portion. The Young's modulus of the adjusting component is greater than the Young's modulus of the first elastic portion. The first elastic portion and the adjusting component have different metal materials.

(20) According to some embodiments of the present disclosure, the first elastic portion includes a first opening and a second opening. The movable portion further includes a third stopper portion passes through the first opening and the second opening respectively. The first opening and the

second opening have independent structures. When the movable portion is in a third limiting position, the third stopper portion directly or indirectly contacts the third sidewall of the fixed portion.

(21) According to some embodiments of the present disclosure, the tactile feedback mechanism further includes a third buffer component. The third buffer component is fixedly disposed on the third stopper portion or the third sidewall of the fixed portion. The third buffer component has a plastic material.

(22) According to some embodiments of the present disclosure, the driving assembly includes a coil, a magnetic assembly, and a magnetic permeability component. The magnetic assembly corresponds to the coil. The magnetic permeability component is fixedly disposed on the magnetic assembly and has a magnetic permeability material. The magnetic permeability component is used to adjust the distribution of the magnetic field generated by the magnetic assembly. The coil and the magnetic assembly are arranged along the direction of the main axis. The movable portion and the coil are at least partially overlapped when viewed along a direction perpendicular to the direction of the main axis.

(23) According to some embodiments of the present disclosure, the magnetic assembly further includes a first magnetic component, a second magnetic component, and a third magnetic component. The second magnetic component and the third magnetic component correspond to the first magnetic component. The first magnetic component, the second magnetic component, and the third magnetic component are arranged along a first axis. The first magnetic component has a first magnetic pole pair including an N pole and an S pole. The second magnetic component has a second magnetic pole pair including an N pole and an S pole. The third magnetic component has a third magnetic pole pair including an N pole and an S pole. The first magnetic pole pair, the second magnetic pole pair, and the third magnetic pole pair are arranged in different orientations respectively. The magnetic pole pairs of the second magnetic pole pair and the third magnetic pole pair are arranged along the first axis. The N pole of the second magnetic pole pair is closer to the first magnetic component than the S pole of the second magnetic pole pair. The N pole of the third magnetic pole pair is closer to the first magnetic component than the S pole of the third magnetic pole pair. The S pole of the first magnetic pole pair is closer to the magnetic permeability component than the N pole of the first magnetic pole pair. The N pole of the first magnetic pole pair is closer to the coil than the S pole of the first magnetic pole pair.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to make the above and other objectives, features, and advantages of the present disclosure more obvious and understandable, preferred embodiments are listed below in conjunction with the accompanying drawings, which are described in detail as follows.

(2) FIG. **1** is an exploded view of the tactile feedback mechanism according to some embodiments of the disclosure.

(3) FIG. **2**A is a perspective view of a portion of the tactile feedback mechanism according to some embodiments of the present disclosure, showing the movable portion, the housing, the connecting assembly, and the magnetic assembly.

(4) FIG. **2**B is a front view of the portion of the tactile feedback mechanism of FIG. **2**A according to some embodiments of the disclosure, showing the movable portion, the housing, the connecting assembly, and the magnetic assembly.

(5) FIG. **3**A is a partial front view showing the tactile feedback mechanism in the first limiting position according to some embodiments of the present disclosure.

(6) FIG. **3**B is another partial front view of the tactile feedback mechanism in the first limiting

position according to some embodiments of the present disclosure, including the first buffer component.
(7) FIG. **3**C is a partial front view of the tactile feedback mechanism in the second limiting position according to some embodiments of the present disclosure.
(8) FIG. **3**D is another partial front view of the tactile feedback mechanism in the second limiting position according to some embodiments of the present disclosure, including the second buffer component.
(9) FIG. **4**A is another perspective view of a portion of the tactile feedback mechanism according to some embodiments of the present disclosure, showing the movable portion, the housing, the connecting assembly, and the magnetic assembly.
(10) FIG. **4**B is a front view of the portion of the tactile feedback mechanism of FIG. **4**A according to some embodiments of the disclosure, showing the movable portion, the housing, the connecting assembly, and the magnetic assembly.
(11) FIG. **5**A is another partial front view of the tactile feedback mechanism in the first limiting position according to some embodiments of the present disclosure.
(12) FIG. **5**B is another partial front view of the tactile feedback mechanism in the first limiting position according to some embodiments of the present disclosure, including another first buffer component.
(13) FIG. **6**A is a perspective view of a portion of the tactile feedback mechanism according to some embodiments of the present disclosure, showing the movable portion, the connecting assembly, the magnetic assembly, and the adjusting component.
(14) FIG. **6**B is a perspective view of a portion of the tactile feedback mechanism according to some embodiments of the present disclosure, showing the movable portion, the connecting assembly, and the magnetic assembly, wherein the connecting assembly includes the first opening and the second opening.
(15) FIG. **6**C is a perspective view of a portion of the tactile feedback mechanism according to some embodiments of the present disclosure, showing the movable portion, the connecting assembly, and the magnetic assembly, wherein the movable portion includes the third stopper portion.
(16) FIG. **7**A is a partial front view of the tactile feedback mechanism in the third limiting position according to some embodiments of the present disclosure.
(17) FIG. **7**B is another partial front view of the tactile feedback mechanism in the third limiting position according to some embodiments of the present disclosure, including the third buffer component.
(18) FIG. **8**A is an exploded view of the movable portion and the magnetic assembly according to some embodiments of the disclosure.
(19) FIG. **8**B is a cross-sectional view of FIG. **8**A taken along line F according to some embodiments of the present disclosure.
(20) FIG. **9** is a cross-sectional side view of the tactile feedback mechanism according to some embodiments of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(21) In order to make the purpose, features, and advantages of the present disclosure more obvious and understandable, the following embodiments are specially cited, and the accompanying drawings are used for detailed description. The exemplary embodiments set forth herein are used merely for the purpose of illustration, and the inventive concept can be embodied in various forms without being limited to those exemplary embodiments. In addition, the drawings of different embodiments can use like and/or corresponding numerals to denote like and/or corresponding components in order to clearly describe the present disclosure. However, the use of like and/or corresponding numerals in the drawings of different embodiments does not suggest any correlation between different embodiments. The directional terms, such as “up”, “down”, “left”, “right”,

"front" or "rear", are reference directions for accompanying drawings. Therefore, using the directional terms is for description instead of limiting the disclosure.

(22) In addition, relative terms such as "lower" or "bottom" and "upper" or "top" may be used in the embodiments to describe the relative relationship between one component and another component of the illustration. It should be understood that if the illustrated device is turned upside down, the components described on the "lower" side will become the components on the "upper" side.

(23) The making and using of the embodiments of the tactile feedback mechanism are discussed in detail below. It should be appreciated, however, that the embodiments provide many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use the embodiments, and do not limit the scope of the disclosure. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this invention belongs. It should be appreciated that each term, which is defined in a commonly used dictionary, should be interpreted as having a meaning conforming to the relative skills and the background or the context of the present disclosure, and should not be interpreted in an idealized or overly formal manner unless defined otherwise.

(24) Referring to FIG. **1**, which is an exploded view of the tactile feedback mechanism **10** according to some embodiments of the present disclosure. As shown in FIG. **1**, the tactile feedback mechanism **10** includes a movable portion **100**, a fixed portion **200**, a driving assembly **300**, and a connecting assembly **400**A. The fixed portion **200** includes a housing **210** and a base **220**. The driving assembly **300** includes a coil **310**, a set of magnetic assembly **320**, a magnetic permeability component **330**, and a printed circuit board **340**.

(25) The movable portion **100** has an upper portion **150** and a lower portion **160**. The upper portion **150** of the movable portion **100** can be seen in FIG. **1**, and the lower portion **160** of the movable portion **100** cannot be seen in the perspective of FIG. **1**. The housing **210** of the fixed portion **200** has a top surface **215**, a first sidewall **211**, a second sidewall **212**, a third sidewall **213**, and a fourth sidewall **214**. The top surface **215**, the second sidewall **212**, and the third sidewall **213** of the housing **210** can be seen in FIG. **1**.

(26) The housing **210** and the base **220** of the fixed portion **200** can be combined to form a space for accommodating the parts inside the fixed portion **200**. The driving assembly **300** can drive the movable portion **100** to vibrate in the direction of a first axis D**1** relative to the fixed portion **200**. The connecting assembly **400**A is flexible. The movable portion **100** is movably connected to the fixed portion **200** via the connecting assembly **400**A.

(27) The magnetic assembly **320** of the driving assembly **300** corresponds to the coil **310**. The magnetic permeability component **330** is fixedly disposed on the magnetic assembly **320** and has a magnetic permeability material. The magnetic permeability component **330** is used to adjust the distribution of the magnetic field generated by the magnetic assembly **320**. The coil **310** and the magnetic assembly **320** are arranged along the direction of a main axis D**3**. When viewed along a direction perpendicular to the main axis D**3** (for example, the first axis D**1** or a second axis D**2**), the movable portion **100** and the coil **310** at least partially overlap. This arrangement may reduce the thickness of the tactile feedback mechanism **10** in the direction of the main axis D**3**, to achieve the effect of miniaturization.

(28) FIG. **2**A shows a perspective view of a portion of the tactile feedback mechanism **10** according to some embodiments of the present disclosure. The lower portion **160** of the movable portion **100**, the second sidewall **212** and the third sidewall **213** of the housing **210**, the magnetic assembly **320**, and the connecting assembly **400**A can be seen. The lower portion **160** of the movable portion **100** also includes a recess portion **165** that may accommodate the coil **310** (FIG. **1**) upon assemble.

(29) FIG. **2**B is a front view of the portion of the tactile feedback mechanism **10** shown in FIG. **2**A in the initial position. The lower portion **160** of the movable portion **100**, the first sidewall **211**, the

second sidewall **212**, the third sidewall **213**, and the fourth sidewall **214** of the housing **210** can be seen. The magnetic assembly **320** and the connecting assembly **400**A can also be seen.
(30) In the embodiment shown in FIG. **2**B, the connecting assembly **400**A has a first fixed end for the movable portion **410**A, a second fixed end for the movable portion **415**A, a first fixed end for the fixed portion **420**A, a first elastic portion **430**A, and a second elastic portion **435**A. The first fixed end for the movable portion **410**A and the second fixed end for the movable portion **415**A are fixedly connected to the movable portion **100**. The first fixed end for the fixed portion **420**A is fixedly connected to the first sidewall **211** of the housing **210**.
(31) The movable portion **100** and the connecting assembly **400**A are made of different metal materials. The density of the metal material of the connecting assembly **400**A is lower than the density of the metal material of the movable portion **100**. For example, the connecting assembly **400**A may be made of copper alloy, and the movable portion **100** may be made of tungsten. This configuration may increase the vibration intensity of the movable portion **100**, thereby improving the performance of the tactile feedback mechanism **10**.
(32) The first fixed end for the movable portion **410**A, the second fixed end for the movable portion **415**A, and the first fixed end for the fixed portion **420**A are formed in one piece. The first fixed end for the movable portion **410**A is connected to the first fixed end for the fixed portion **420**A via the first elastic portion **430**A. The second fixed end for the movable portion **415**A is connected to the first fixed end for the fixed portion **420**A via the second elastic portion **435**A. Both the first elastic portion **430**A and the second elastic portion **435**A are flexible.
(33) The first fixed end for the movable portion **410**A and the second fixed end for the movable portion **415**A are arranged along the direction of the first axis D**1** and are parallel to each other. This arrangement makes the laser welding process during assembly easier to perform.
(34) The movable portion **100** includes a first connecting portion **105**, a second connecting portion **125**, a first stopper portion **110**, and a second stopper portion **120**. The first connecting portion **105** includes a first connecting surface **1055**, and the second connecting portion **125** includes a second connecting surface **1255**. Both the first connecting surface **1055** and the second connecting surface **1255** face the second sidewall **212** of the housing **210**.
(35) It should be noted that in the initial position, the first stopper portion **110** of the movable portion **100** does not contact the first fixed end for the fixed portion **420**A, and the second stopper portion **120** of the movable portion **100** does not contact the second sidewall **212** of the housing **210**.
(36) The first connecting portion **105** and the second connecting portion **125** both have a recessed structure relative to the second stopper portion **120**. In other words, relative to the protruding structure of the second stopper portion **120**, the first connecting portion **105** and the second connecting portion **125** are in a recessed portion, and wherein the first fixed end for the movable portion **410**A is fixed to the first connecting surface **1055** of the first connecting portion **105**, the second fixed end for the movable portion **415**A is fixed to the second connecting surface **1255** of the second connecting portion **125**. This configuration enables the tactile feedback mechanism **10** of the present disclosure to achieve miniaturization.
(37) The first stopper portion **110** of the movable portion **100** is located close to the first sidewall **211** of the housing **210**. The second stopper portion **120** of the movable portion **100** is located close to the second sidewall **212** of the housing **210**. FIGS. **3**A to **3**D show the partial front views of the two frames A and C in FIG. **2**B, with the first stopper portion **110** in the first limiting position and the second stopper portion **120** in the second limiting position respectively.
(38) According to some embodiments of the present disclosure, FIGS. **3**A and **3**B show partial front views of the movable portion **100** in the frame A of FIG. **2**B in the first limiting position. When the movable portion **100** is in the first limiting position, the first stopper portion **110** limits the range of movement of the movable portion **100** relative to the fixed portion **200** (FIG. **1**). As shown in FIGS. **3**A and **3**B, in the first limiting position, the first stopper portion **110** can directly or

indirectly contact the first fixed end for the fixed portion **420**A.

(39) In the embodiment shown in FIG. **3**A, the movable portion **100** directly contacts the first fixed end for the fixed portion **420**A in the first limiting position. In the embodiment shown in FIG. **3**B, the tactile feedback mechanism **10** further includes a first buffer component **610**A. The first buffer component **610**A is made of a plastic material, and may be arranged on the first stopper portion **110** or the first fixed end for the fixed portion **420**A. When the movable portion **100** is in the first limiting position, the first stopper portion **110** indirectly contacts the first fixed end for the fixed portion **420**A via the first buffer component **610**A, so that the movable portion **100** will not be damaged in the first limiting position.

(40) FIGS. **3**C and **3**D show partial front views of the movable portion **100** in the frame C of FIG. **2**B in the second limiting position. In the embodiment shown in FIG. **3**C, the second stopper portion **120** is located between the first connecting portion **105** and the second connecting portion **125** when viewed along the direction of the main axis D**3**.

(41) Referring to FIG. **3**C, when the movable portion **100** is in the second limiting position, the second stopper portion **120** limits the range of movement of the movable portion **100** relative to the fixed portion **200** (FIG. **1**). As shown in FIGS. **3**C and **3**D, the second stopper portion **120** can directly or indirectly contact the second sidewall **212** of the housing **210** in the second limiting position.

(42) In the embodiment shown in FIG. **3**C, the second stopper portion **120** of the movable portion **100** directly contacts the second sidewall **212** of the housing **210** in the second limiting position. In the embodiment shown in FIG. **3**D, the tactile feedback mechanism **10** further includes a second buffer component **620**. The second buffer component **620** is made of plastic material, and may be disposed on the second stopper portion **120** or the second sidewall **212** of the housing **210**. When the movable portion **100** is in the second limiting position, the second stopper portion **120** indirectly contacts the second sidewall **212** of the housing **210** via the second buffer component **620**.

(43) FIG. **4**A shows a perspective view of a portion of the tactile feedback mechanism **10** according to another embodiment of the present disclosure. In FIG. **4**A, the lower portion **160** of the movable portion **100**, the housing **210**, the magnetic assembly **320**, and a connecting assembly **400**B can be seen. The tactile feedback mechanism **10** of FIG. **4**A is substantially similar to the tactile feedback mechanism **10** of FIG. **2**A, and the connecting assembly **400**B is similar to the connecting assembly **400**A described above, and has common features and functions, the details of which will be describe below.

(44) FIG. **4**B is a front view of the portion of the tactile feedback mechanism **10** shown in FIG. **4**A. The lower portion **160** of the movable portion **100**, the first sidewall **211**, the second sidewall **212**, the third sidewall **213**, and the fourth sidewall **214** of the housing **210** can be seen. The magnetic assembly **320** and the connecting assembly **400**B can also be seen. The driving assembly **300** (FIG. **1**) can drive the movable portion **100** to move relative to the fixed portion **200** (FIG. **1**) in the direction of the first axis D**1**. The movable portion **100** is movably connected to the fixed portion **200** (FIG. **1**) via the connecting assembly **400**B, wherein the connecting assembly **400**B is made of a metal material.

(45) In the embodiment shown in FIG. **4**B, the connecting assembly **400**B has a first fixed end for the movable portion **410**B, a first fixed end for the fixed portion **420**B, a second fixed end for the fixed portion **425**B, a first elastic portion **430**B, and a second elastic portion **435**B. The first fixed end for the movable portion **410**B is fixedly connected to the movable portion **100**. The first fixed end for the fixed portion **420**B and the second fixed end for the fixed portion **425**B are fixedly connected to the first sidewall **211** of the fixed portion **200**.

(46) As shown in FIG. **4**B, the first fixed end for the movable portion **410**B is fixedly connected to the movable portion **100**. The first fixed end for the fixed portion **420**B and the second fixed end for the fixed portion **425**B are fixedly connected to the first sidewall **211** of the housing **210**. The

first fixed end for the movable portion **410**B is connected to the first fixed end for the fixed portion **420**B via the first elastic portion **430**B. The first fixed end for the movable portion **410**B is connected to the second fixed end for the fixed portion **425**B via the second elastic portion **435**B. The first elastic portion **430**B and the second elastic portion **435**B are similar to the first elastic portion **430**A and the second elastic portion **435**A shown in FIG. **2**B, and are both flexible.
(47) Continuing to refer to FIG. **4**B, the first fixed end for the movable portion **410**B, the first fixed end for the fixed portion **420**B, and the second fixed end for the fixed portion **425**B are formed in one piece. The first fixed end for the fixed portion **420**B and the second fixed end for the fixed portion **425**B are arranged along the direction of the first axis D**1**.
(48) The first fixed end for the fixed portion **420**B and the second fixed end for the fixed portion **425**B are disposed on the first sidewall **211** of the housing **210**. The first fixed end for the movable portion **410**B, the first fixed end for the fixed portion **420**B, and the second fixed end for the fixed portion **425**B have a plate-shaped structure. The first fixed end for the movable portion **410**B is located between the movable portion **100** and the first sidewall **211**.
(49) When viewed along the direction of the main axis D**3**, the first elastic portion **430**B and the second elastic portion **435**B each have a pointed shape, and the first elastic portion **430**B and the second elastic portion **435**B are mirror-symmetrical with each other relative to the movable portion **100**. This design of folding the connecting assembly **400**B in the limited accommodating space of the tactile feedback mechanism **10** enables the connecting assembly **400**B in this embodiment to have a smaller elastic coefficient, so that the vibration intensity of the tactile feedback mechanism **10** may increase, thereby increasing the performance of the tactile feedback mechanism **10**.
(50) In the second limiting position, the second stopper portion **120** of the embodiment shown in FIG. **4**B operates in the same manner as the second stopper portion **120** in FIGS. **3**C and **3**D. That is to say, when the movable portion **100** in FIG. **4**B is in the second limiting position as shown in FIGS. **3**C and **3**D, the second stopper portion **120** can directly contact the second sidewall **212** as shown in FIG. **3**C, or indirectly contact the second sidewall **212** via the second buffer component **620** provided on the second stopper portion **120** or the second sidewall **212** as shown in FIG. **3**D, to limit the movement range of the movable portion **100**.
(51) According to some embodiments of the present disclosure, FIGS. **5**A and **5**B show partial front views of the movable portion **100** in the frame D in FIG. **4**B in the first limiting position. When the movable portion **100** is in the first limiting position, the first fixed end for the movable portion **410**B limits the range of movement of the movable portion **100** relative to the fixed portion **200** (FIG. **1**). As shown in FIGS. **5**A and **5**B, in the first limiting position, the first fixed end for the movable portion **410**B may directly or indirectly contact the first fixed end for the fixed portion **420**B and the second fixed end for the fixed portion **425**B.
(52) In the embodiment shown in FIG. **5**A, the first fixed end for the movable portion **410**B directly contacts the first fixed end for the fixed portion **420**B and the second fixed end for the fixed portion **425**B in the first limiting position. In the embodiment shown in FIG. **5**B, the tactile feedback mechanism **10** further includes a first buffer component **610**B. The first buffer component **610**B is made of plastic material, and may be disposed on the first fixed end for the movable portion **410**B or both the first fixed end for the fixed portion **420**B and the second fixed end for the fixed portion **425**B. When the movable portion **100** is in the first limiting position, the first fixed end for the movable portion **410**B indirectly contacts the first fixed end for the fixed portion **420**B and the second fixed end for the fixed portion **425**B via the first buffer component **610**B.
(53) According to some embodiments of the present disclosure, FIG. **6**A shows a perspective view of the movable portion **100**, the magnetic assembly **320**, the connecting assembly **400**B, and an adjusting component **700** of the tactile feedback mechanism **10**. In the embodiment shown in FIG. **6**A, the adjusting component **700** is disposed on the first elastic portion **430**B and the second elastic portion **435**B of the connecting assembly **400**B.
(54) As shown in FIG. **6**A, the adjusting component **700** has a plate-like structure. The Young's

modulus of the adjusting component **700** is greater than the Young's modulus of the first elastic portion **430**B and the second elastic portion **435**B. The adjusting component **700** may be used to adjust the elastic coefficients of the first elastic portion **430**B and the second elastic portion **435**B, to change the vibration intensity and vibration frequency of the tactile feedback mechanism **10**. The adjusting component **700** and the connecting assembly **400**B are made of different metal materials, for example, different copper alloys.
(55) According to some embodiments of the present disclosure, FIG. **6**B shows a perspective view of the movable portion **100**, the magnetic assembly **320**, and the connecting assembly **400**B. In the embodiment shown in FIG. **6**B, the first elastic portion **430**B and the second elastic portion **435**B each have a first opening **431** and a second opening **432**, respectively. The first opening **431** and the second opening **432** are independent structures formed on the first elastic portion **430**B and the second elastic portion **435**B. That is to say, the first elastic portion **430**B and the second elastic portion **435**B each have a first opening **431** and a second opening **432**, and the structure of the first opening **431** is not affected by the structure of the second opening **432**, and vice versa. This arrangement may reduce the elastic coefficient of the connecting assembly **400**B, so as to increase the vibration intensity of the tactile feedback mechanism **10**, thereby improving the performance of the tactile feedback mechanism **10**.
(56) According to some embodiments of the present disclosure, FIG. **6**C shows a perspective view of the movable portion **100**, the magnetic assembly **320**, and the connecting assembly **400**B. In the embodiment shown in FIG. **6**C, the movable portion **100** includes two third stopper portions **130**, and the first elastic portion **430**B and the second elastic portion **435**B each have a first opening **431** and a second opening **432**.
(57) As shown in FIG. **6**C, the third stopper portion **130** is a protruding structure on the two surfaces of the movable portion **100** near the third sidewall **213** (not shown in FIG. **6**C) and the fourth sidewall **214** (not shown in FIG. **6**C). The first openings **431** and the second openings **432** are formed on the first elastic portion **430**B and the second elastic portion **435**B as shown in FIG. **6**B. The two third stopper portions **130** respectively pass through the first openings **431** and the second openings **432** on the first elastic part **430**B and the second elastic part **435**B.
(58) This configuration not only reduces the elastic coefficient of the connecting assembly **400**B, but also increases the mass of the movable portion **100**, which may increase the vibration intensity and the maximum acceleration of the tactile feedback mechanism **10**, thereby improving the performance of the tactile feedback mechanism **10**.
(59) According to some embodiments of the present disclosure, FIGS. **7**A and **7**B show partial front views of the movable portion **100** in the frame E of FIG. **6**C in the third limiting position. When the movable portion **100** is in the third limiting position, the third stopper portion **130** limits the range of movement of the movable portion **100** relative to the fixed portion **200** (not shown). As shown in FIGS. **7**A and **7**B, in the third limiting position, the third stopper portion **130** may directly or indirectly contact the third sidewall **213** of the housing **210**.
(60) In the embodiment shown in FIG. **7**A, the third stopper portion **130** directly contacts the third sidewall **213** in the third limiting position. The embodiment shown in FIG. **7**B further includes a third buffer component **630**. The third buffer component **630** is made of plastic material, and may be disposed on the third stopper portion **130** or the third sidewall **213**. When the movable portion **100** is in the third limiting position, the third stopper portion **130** indirectly contacts the third sidewall **213** via the third buffer component **630**.
(61) According to some embodiments of the present disclosure, another third stopper portion **130** (shown in FIG. **6**C) close to the fourth sidewall **214** (shown in FIG. **4**B) may also directly or indirectly contact the fourth sidewall **214** via another buffer element in a fourth limiting position in a similar manner as described above.
(62) Referring to FIG. **8**A, the magnetic assembly **320** of the present disclosure may also be a Halbach array. In FIG. **8**A, an exploded view of the lower portion **160** of the movable portion **100**

and the magnetic assembly **320** can be seen. In this embodiment, the magnetic assembly **320** includes a first magnetic component **321**, a second magnetic component **322**, a third magnetic component **323**, a fourth magnetic component **324**, and a fifth magnetic component **325**. The arrangement of the magnetic assembly **320** in a Halbach array will be described in detail below.

(63) FIG. **8**B shows a cross-sectional view of FIG. **8**A taken along line F. The movable portion **100** and the magnetic assembly **320** can be seen. Hereinafter, the first magnetic component **321**, the second magnetic component **322**, the third magnetic component **323**, and the fourth magnetic component **324** are used as examples to illustrate the arrangement of a Halbach array. The first magnetic component **321** has a first magnetic pole pair including an N pole and an S pole. The second magnetic component **322** has a second magnetic pole pair including an N pole and an S pole. The third magnetic component **323** has a third magnetic pole pair including an N pole and an S pole. The fourth magnetic component **324** has a fourth magnetic pole pair including an N pole and an S pole.

(64) The first magnetic pole pair, the second magnetic pole pair, the third magnetic pole pair, and the fourth magnetic pole pair are arranged in different orientations, respectively. The magnetic poles of the second magnetic pole pair and the third magnetic pole pair are arranged along the direction of the first axis D**1**, and the magnetic poles of the first magnetic pole pair and the fourth magnetic pole pair are arranged along the direction of the main axis D**3**.

(65) The N pole of the second magnetic pole pair is closer to the first magnetic component **321** than the S pole of the second magnetic pole pair, and the N pole of the third magnetic pole pair is closer to the first magnetic component **321** than the S pole of the third magnetic pole pair. The S pole of the first magnetic pole pair is closer to the magnetic permeability component **330** than the N pole of the first magnetic pole pair, and the N pole of the first magnetic pole pair is closer to the recess portion **165** that may accommodate the coil **310** than the S pole of the first magnetic pole pair. The S pole of the fourth magnetic pole pair is closer to the recess portion **165** that may accommodate the coil **310** than the N pole of the fourth magnetic pole pair, and the N pole of the fourth magnetic pole pair is closer to the magnetic permeability component **330** than the S pole of the fourth magnetic pole pair.

(66) In this configuration, the magnetic field of the magnetic assembly **320** on the side of the recess portion **165** that can accommodate the coil **310** may be significantly enhanced, and the generated magnetic field is shown by the arrow near the recess portion **165** in FIG. **8**B. With this arrangement, the number of coils required may be reduced, and the effect of miniaturization may be achieved. This arrangement may also generate a stronger magnetic field with less magnetic components, which would normally require more of the general magnetic assembly **320** to achieve, thereby improving the performance.

(67) According to some embodiments of the present disclosure, FIG. **9** shows a cross-sectional side view of the tactile feedback mechanism **10**. The movable portion **100**, the housing **210**, the base **220**, the coil **310**, the magnetic assembly **320**, the printed circuit board **340**, and the connecting assembly **400** can be seen. In some embodiments, the tactile feedback mechanism **10** may have a damping component **500** in the non-moving axial direction (for example, the direction of the main axis D**3** in this embodiment). The damping component **500** may be located between the movable portion **100** and the fixed portion **200** as shown in FIG. **9**. The addition of the damping component **500** may adjust the motion parameters of the tactile feedback mechanism **10**, so that the disclosed device may increase stability and reduce noise.

(68) The present disclosure provides a tactile feedback mechanism. The movable portion is movably connected to the fixed portion via the connecting assembly. The driving assembly drives the movable portion to move relative to the fixed portion to vibrate. The present disclosure achieves the effects of miniaturization and easy assembly by the different configurations of the movable portion and the connecting assembly. The present disclosure also increases the vibration intensity and maximum acceleration of the tactile feedback mechanism by regulating the elastic

coefficient of the connecting assembly and increasing the mass of the movable portion, thereby improving the stability and performance of the tactile feedback mechanism.
(69) Although the embodiments and their advantages have been described in detail, it should be understood that various changes, substitutions, and alterations can be made herein without departing from the spirit and scope of the embodiments as defined by the appended claims. Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods, and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein can be utilized according to the disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps. In addition, each claim constitutes a separate embodiment, and the combination of various claims and embodiments are within the scope of the disclosure.
(70) It will be apparent to those skilled in the art that various modifications and variations can be made in the invention. It is intended that the standard and examples be considered as exemplary only, with the true scope of the disclosed embodiments being indicated by the following claims and their equivalents.

Claims

1. A tactile feedback mechanism, comprising: a movable portion; a fixed portion, wherein the movable portion is movable relative to the fixed portion; a driving assembly configured to drive the movable portion to move relative to the fixed portion; and a connecting assembly, wherein the movable portion is movably connected to the fixed portion via the connecting assembly, and the connecting assembly has a metal material; wherein the connecting assembly comprises: a first fixed end for the movable portion, fixedly connected to the movable portion; a first fixed end for the fixed portion, fixedly connected to the fixed portion; a first elastic portion, wherein the first fixed end for the movable portion is movably connected to the first fixed end for the fixed portion via the first elastic portion; a second fixed end for the fixed portion, fixedly connected to the fixed portion; and a second elastic portion, wherein the first fixed end for the movable portion is connected to the second fixed end for the fixed portion via the second elastic portion; wherein the first fixed end for the fixed portion is connected to the second fixed end for the fixed portion via the first fixed end for the movable portion, and the first fixed end for the movable portion, the first fixed end for the fixed portion, and the second fixed end for the fixed portion are formed in one piece, the first fixed end for the fixed portion and the second fixed end for the fixed portion are arranged along a first axis; wherein the fixed portion comprises a first sidewall, the first fixed end for the fixed portion and the second fixed end for the fixed portion are disposed on the first sidewall, the first fixed end for the movable portion, the first fixed end for the fixed portion, the second fixed end for the fixed portion, and the first sidewall have plate-shaped structures, and the first fixed end for the movable portion is located between the movable portion and the first sidewall.
2. The tactile feedback mechanism as claimed in claim 1, wherein the first elastic portion has a metal material; wherein the movable portion has a metal material, and the metal material of the first elastic portion is different from the metal material of the movable portion, and the density of the metal material of the first elastic portion is lower than the density of the metal material of the movable portion.
3. The tactile feedback mechanism as claimed in claim 2, wherein the driving assembly comprises: a coil; a magnetic assembly, corresponding to the coil; a magnetic permeability component, fixedly disposed on the magnetic assembly and having a magnetic permeability material; wherein the

magnetic permeability component is used to adjust the distribution of the magnetic field generated by the magnetic assembly, and the coil and the magnetic assembly are arranged along a direction of a main axis, the movable portion and the coil are at least partially overlapped when viewed along a direction perpendicular to the direction of the main axis.

4. The tactile feedback mechanism as claimed in claim 3, wherein the magnetic assembly further comprises a first magnetic component, a second magnetic component, and a third magnetic component, wherein the second magnetic component and the third magnetic component correspond to the first magnetic component, the first magnetic component, the second magnetic component, and the third magnetic component are arranged along the first axis, wherein the first magnetic component has a first magnetic pole pair comprising an N pole and an S pole, and the second magnetic component has a second magnetic pole pair comprising an N pole and an S pole, the third magnetic component has a third magnetic pole pair comprising an N pole and an S pole, the first magnetic pole pair, the second magnetic pole pair, and the third magnetic pole pair are arranged in different orientations respectively, the magnetic pole pairs of the second magnetic pole pair and the third magnetic pole pair are arranged along the first axis, and the N pole of the second magnetic pole pair is closer to the first magnetic component than the S pole of the second magnetic pole pair, the N pole of the third magnetic pole pair is closer to the first magnetic component than the S pole of the third magnetic pole pair, and the S pole of the first magnetic pole pair is closer to the magnetic permeability component than the N pole of the first magnetic pole pair, the N pole of the first magnetic pole pair is closer to the coil than the S pole of the first magnetic pole pair.

5. The tactile feedback mechanism as claimed in claim 1, wherein the connecting assembly further comprises: a second fixed end for the movable portion, fixedly connected to the movable portion; wherein the second fixed end for the movable portion is connected to the first fixed end for the fixed portion via the second elastic portion; wherein the first fixed end for the movable portion is connected to the second fixed end for the movable portion via the first fixed end for the fixed portion, and the first fixed end for the movable portion, the first fixed end for the fixed portion, and the second fixed end for the movable portion are formed in one piece.

6. The tactile feedback mechanism as claimed in claim 5, wherein the first fixed end for the fixed portion is disposed on the first sidewall, and the movable portion comprises a first stopper portion for limiting the range of movement of the movable portion relative to the fixed portion, the first stopper portion directly or indirectly contacts the first fixed end for the fixed portion when the movable portion is in a first limiting position.

7. The tactile feedback mechanism as claimed in claim 6, further comprising a first buffer component, the first buffer component has a plastic material, and the first buffer component is fixedly disposed on the first stopper portion or the first fixed end for the fixed portion.

8. The tactile feedback mechanism as claimed in claim 6, wherein the movable portion comprises a first connecting portion and a second connecting portion, the first fixed end for the movable portion is fixed to the first connecting portion, and the second fixed end for the movable portion is fixed to the second connecting portion, wherein the first connecting portion and the second connecting portion both have a recessed structure.

9. The tactile feedback mechanism as claimed in claim 8, wherein the first connecting portion comprises a first connecting surface, the second connecting portion comprises a second connecting surface, the first fixed end for the movable portion is disposed on the first connecting surface, the second fixed end for the movable portion is disposed on the second connecting surface, the fixed portion comprises a second sidewall, the first fixed end for the movable portion and the second fixed end for the movable portion are arranged along the first axis, and the first connecting surface and the second connecting surface face the second sidewall.

10. The tactile feedback mechanism as claimed in claim 5, further comprising a damping component, the movable portion is movably connected to the fixed portion via the damping component, wherein the damping component is flexible, and the damping component has a plastic

material, the elastic coefficient of the damping component is lower than the elastic coefficient of the first elastic portion.

11. The tactile feedback mechanism as claimed in claim 5, wherein the movable portion comprises a second stopper portion, for limiting the range of movement of the movable portion relative to the fixed portion, and the second stopper portion directly or indirectly contacts the second sidewall of the fixed portion when the movable portion is in a second limiting position.

12. The tactile feedback mechanism as claimed in claim 11, further comprising a second buffer component, wherein the second buffer component has a plastic material, and the second buffer component is fixedly disposed on the second stopper portion or the second sidewall.

13. The tactile feedback mechanism as claimed in claim 1, wherein when the first fixed end for the movable portion directly or indirectly contacts both the first fixed end for the fixed portion and the second fixed end for the fixed portion, the range of movement of the movable portion relative to the fixed portion is limited.

14. The tactile feedback mechanism as claimed in claim 13, further comprising a first buffer component, the first buffer component has a plastic material, and the first buffer component is fixedly disposed on the first fixed end for the movable portion or both the first fixed end for the fixed portion and the second fixed end for the fixed portion.

15. The tactile feedback mechanism as claimed in claim 13, wherein the first elastic portion and the second elastic portion each have a pointed shape when viewed along a direction of a main axis, and the first elastic portion and the second elastic portion are mirror-symmetrical with each other relative to the movable portion.

16. The tactile feedback mechanism as claimed in claim 15, wherein the connecting assembly further comprises an adjusting component, the adjusting component has a plate-like structure and is disposed on the first elastic portion for adjusting the elastic coefficient of the first elastic portion, the Young's modulus of the adjusting component is greater than the Young's modulus of the first elastic portion, and the first elastic portion and the adjusting component have different metal materials.

17. The tactile feedback mechanism as claimed in claim 15, wherein the first elastic portion comprises a first opening and a second opening, and the movable portion further comprises a third stopper portion passing through the first opening and the second opening, wherein the first opening and the second opening have independent structures, and the third stopper portion directly or indirectly contacts the third sidewall of the fixed portion when the movable portion is in a third limiting position.

18. The tactile feedback mechanism as claimed in claim 17, further comprising a third buffer component, the third buffer component is fixedly disposed on the third stopper portion or the third sidewall of the fixed portion, and the third buffer component has a plastic material.
